

Unit -1 Software Project Management Concepts [8 Hrs.]

Introduction,

Project and Software project,

Software project vs other project,

Importance and Problems in software project management,

Process of SPM.

Characteristics of good project manager,

Successful Software Project Manager,

Overview of Software Project Planning.

Unit-2 Software Project Scheduling [8 Hrs.]

2.1 Objectives of activity planning,

2. Work breakdown structure (WBS),

2.3 Network planning model:

 Critical path method (CPM),

 Program evaluation and review technique (PERT),

 Precedence diagramming method (PDM),

2.4 Shortening project duration,

2.5 Identifying critical activities.

2.6 Forward pass and Backward pass

Unit -3 Software Estimation Techniques [7 Hrs.]

Software Effort Estimation:

Problems with over and under estimations,

Basis of software Estimating,

Software effort estimation techniques,

- expert Judgment,
- Estimating by analogy.
- Bottoms-up estimating,
- Top-down approach and
- parametric models.

Unit -4 Software Evaluation and Costing [8 Hrs.]

Project Evaluation:

Strategic Assessment,

Technical Assessment,

cost-benefit analysis,

Cash flow forecasting,

cost-benefit evaluation techniques,

Risk Evaluation.

Selection of Appropriate Report,

Project approach:

Choosing technologies,

choice of process models, structured methods.

Unit-5 Risk Management [5 Hrs.]

Risk Identification, Planning, Evaluation and Management,
Categories of Risk,
Framework for dealing with risk,
Evaluating Risks to the schedule.

This unit focuses on the critical discipline of risk management in software project management.

Risk management is the process of identifying, assessing, mitigating, and monitoring risks that can impact a project's success.

This unit provides a comprehensive understanding of how to effectively manage risks throughout the project's lifecycle.

Unit 6: Software Quality Management

Unit 7: Software Configuration Management (SCM)

What is a project?

- Dictionary definition – a **planned activity**
- Being planned assumes that to a large extent we can determine *how we are going to carry out a task before we start.*
- Planning sometimes may not be an essence.
 - for example; **routine maintenance** : might have been performed so many times that everyone involved knows exactly what needs to be done, and thus planning hardly seems necessary but the procedures might need to be documented to ensure consistency and to help newcomers to the job.
- The first time you do a routine task ----- > > it will be like a project.

Characteristics of a Project

- *non routine tasks* are involved
- *planning* is required
- specific *objectives are to be met* or specified product is to be created
- the project has a *pre-determined time* span
- work is carried out for *someone other* than yourself
- work involves several *specialism*
- work is carried out in *several phases*
- the *resources* that are available for use on the project are *constrained*
- the project is *large or complex*

Key Characteristics of Software Project

- **Complexity:** Software projects are often complex, with numerous interdependent components.
- **Changeability:** Software can be easily modified, which leads to evolving requirements and scope changes.
- **Intangibility:** Unlike physical products, software is intangible, which requires careful specification and testing.

What is a Software Project?

- Collection and integration of various components of the software
- Begins with the early investigation and ends with implementation

A project is a temporary endeavor with a specific goal, a defined beginning and end, and allocated resources.

Software projects focus on developing software products or solutions, which involve unique challenges and considerations.

What is management?

- The Open University Software Project management module (1987) suggested that management involves the following activities:
 - *Planning: deciding what is to be done;*
 - *Organizing: making arrangements ;*
 - *Staffing: selecting the right people for the right job;*
 - *Directing: giving instructions;*
 - *Monitoring: checking on progress;*
 - *Controlling: taking action to remedy hold ups;*
 - *Innovating: Coming up with new solutions;*
 - *Representing: coordinate with users;*

Software project versus other types of project

The product of software project have certain characteristics which make them different.

Invisibility: When a physical artifact such a bridge or road is being constructed the progress being made can actually be seen. *With software, progress is not immediately visible.*

Complexity: *software products contain more complexity* than other engineered artifacts.

Conformity: The traditional engineer is usually working *with physical systems and physical materials like cement and steel.* These systems can have some complexity but are governed by physical laws that are consistent. Software developers have to conform to the requirements of human clients *that certainly keeps on fluctuating.*

Flexibility: The ease with which *software can be changed* is usually seen as one of its strengths. Software systems are subject to a high degree of change.

Activities covered by SPM

- A software project is not only concerned with the actual writing of software.
- “Off the shelf software” ???
 - Is still fundamentally a software project as so many of the other elements associated with this type of project is present.
- Three successive processes bring a new system into being.

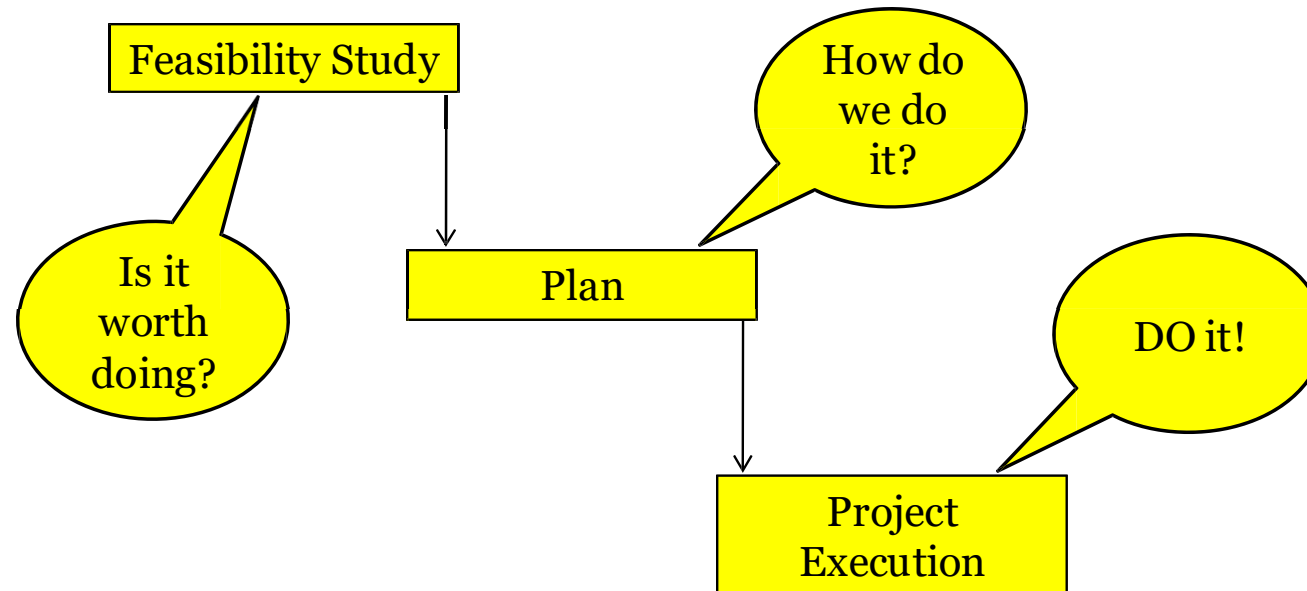


Fig: The feasibility study/plan/execution cycle

Activities covered by SPM (contd...)

The feasibility study:

- Is an investigation into whether a prospective project is worth starting or not ?
- Information gathering (about the proposed application)
- Probable development and operational costs ???
- Estimation of benefits from new system.
- A strategic planning for the following studies
 - ▮ **Technical** – hardware, software infrastructure ???
 - ▮ **Economical** – investment vs rate of return???
 - ▮ **Legal** – legal procedures, IT policy, cyber law, cyber ethics ????
 - ▮ **Operational** – will the system come into operation as per the requirement or not????
 - ▮ **Schedule** – tentative time period of completion

Activities covered by SPM (contd...)

- Contents list for a feasibility study
 - Introduction : identifies what the document is;
 - Description of the current situation;
 - Problem description;
 - Proposed development;
 - Business and financial aspects
 - Technical aspects
 - Organizational aspects;
 - Estimated costs
 - Development costs
 - Operational costs;
 - Envisaged benefits;
 - Recommendation;

Activities covered by SPM (contd...)

- Planning:

- Starts of the feasibility study proves to be viable
- For a large project, detailed planning *is not done right at the beginning.*
- An outline is formulated for the whole project and detailed planning is done for the first stage.
- Detailed planning of later stages is done as they approach
>>>>> this is done because we would have more detailed and accurate information upon which to base our plans nearer to the start of the later stages.

Activities covered by SPM (contd...)

Contents list for a project plan

- Introduction;
- Background: including reference to the business case;
- Project objectives;
- Constraints : these could be included with project objectives
- Project products: both deliverable products that the client will receive and intermediate products;
- Methods;
- Activities to be carried out;
- Resources to be used;
- Risks to the project;
- Management of the project, including
 - Organizational responsibilities
 - Management of quality
 - Configuration of quality

Activities covered by SPM (contd...)

- Project Execution:
 - The project can now be executed.
 - Often contains two subphases
 - ▮ *Design*
 - ▮ *Implementation*
 - Design is *thinking and making decisions* about the precise form of the products that the project is to create.
 - This could be the user interface design or the data flow design or internal architecture design.
 - Implementation refers to the *real time execution* of the final product of the software project.

A typical project life cycle

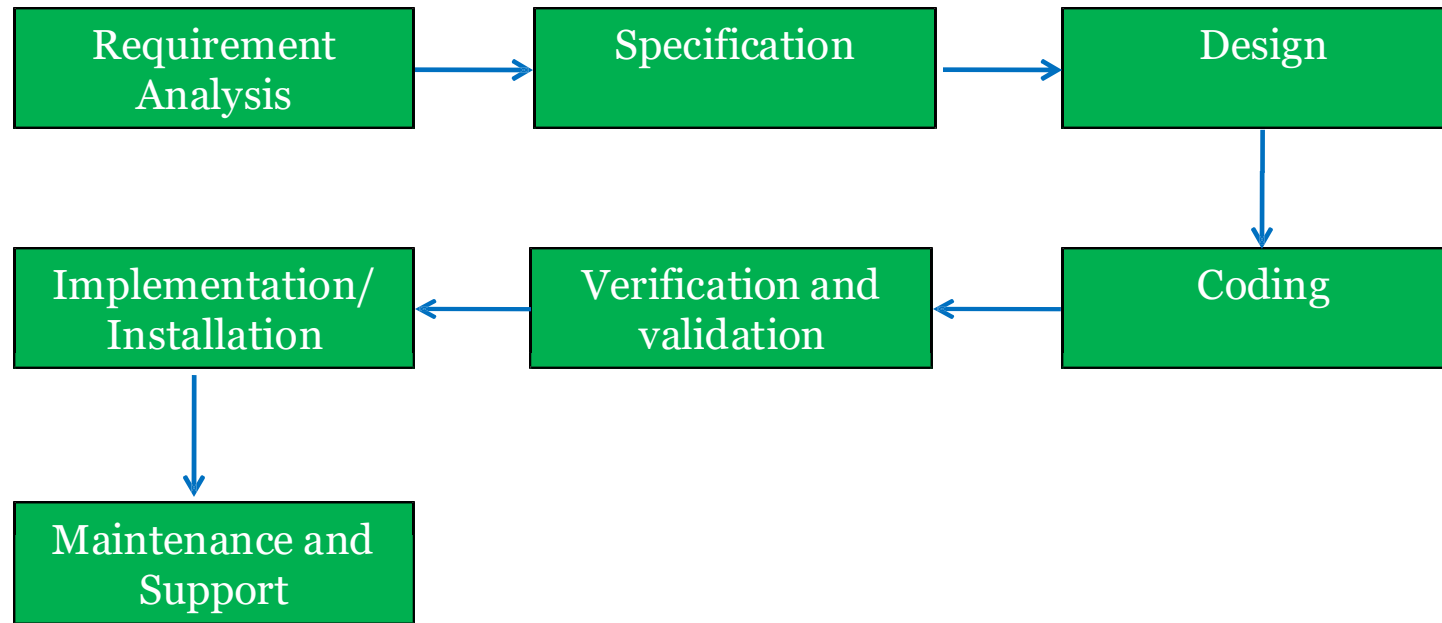


Fig: Project Life Cycle

A typical project life cycle (contd...)

- Requirement Analysis:
 - This is finding out in detail what the users require of the system that the project is to implement. Some works along these lines will almost certainly have been carried out when the project was evaluated, but now the original information obtained needs to be updated and supplemented. Several different approaches to the users' requirements may be explored.
- Specification:
 - Detailed documentation of what the proposed system is to do.
- Design:
 - A design has to be drawn up which meets the specification.
 - Design will be in two stages.
 - One will be the external or user design concerned with the external appearance of the application.
 - The other produces the physical design which tackles the way that the data and software and procedures are to be structured internally.

A typical project life cycle (contd...)

- Coding:
 - This may refer to writing code in a programming language or use any application builder.
 - Even where software is not being built from scratch, some modification to the base package could be required to meet the needs of the new application.
- Verification and Validation:
 - Whether software is developed specially for the current application or not, careful testing will be needed to check that the proposed system meets its requirements.
- Implementation/Installation
 - *Whole of the project after design VS. installation of the system after the software has been developed*
 - The later case includes setting up operational data files and system parameters, writing user manuals and training users of the new system.
- Maintenance and Support
 - Once the system goes live, there is a continuous need for error correction as well as extensions and improvements to the system
 - May be *corrective, adaptive or perfective.*

Categorizing Software Projects

1. INFORMATION SYSTEMS VS. EMBEDDED SYSTEMS

- A distinction may be made between *information systems* and *embedded systems*.

Information System	Embedded System (Real Time / Industrial Systems)
It interfaces with the organization.	It interfaces with a machine.
A stock control system that controls when the organization reorders stock.	A process control system that may have elements of the air conditioning equipment in a building

Would an operating system on a computer be an information system or an embedded system?

Many large organizations that are committed to computer based information systems have specialists responsible for the maintenance of operating systems. However, an operating system is primarily concerned with driving the hardware it is argues that is has more in common with embedded system.

Categorizing Software Projects

2. OBJECTIVES VS. PRODUCTS

- Projects may be distinguished by whether their aim is to produce a product or to meet an objective.

Objectives	Products
The project may be required to meet certain objectives.	A project might be to create a product, the details of which have been specified by the client.
The first stage of a project is always objective driven	The objective driven project a software product.

Problems with Software Project

- One way of deciding what ought to be covered in SPM is to consider what the problems are that it should address.

Commonly experienced problems from manager's point of view

- Poor estimates and plans
- Lack of quality standards and measures
- Lack of guidance about making organizational decisions
- Lack of techniques to make progress visible
- Poor role definition – who does what?
- Incorrect success criteria

Problems with Software Project

*Commonly experienced problems from other stakeholders
(Staff who make up the members of the project team)*

- Inadequate specification of work
- Management ignorance of IT
- Lack of knowledge of application area
- Lack of standard
- Lack of up to date documentation
- Narrow scope of technical expertise
- Changing software environment
- Changing statutory requirements
- Deadline pressure
- Lack of quality control
- Remote management
- Lack of training
- Lack of communication between users and technicians
- Lack of commitment

Problems with Software Project

Commonly experienced problems by the customers

- The US Internal Revenue System was to abandon its tax system modernization programme after having spent \$4 billion.
- The state of California spent \$1 billion on its non functional welfare database system
- The 339 million pound UK air traffic control system was reported as being two years behind schedule.
- In the UK, a Home Office Immigration service computerization project was reported as having missed two deadlines and was nine months late.

Types of Project Plan

Plan	Description
Quality plan	Describes the quality procedures and standards that will be used in a project.
Validation plan	Describes the approach, resources and schedule used for system validation.
Configuration management plan	Describes the configuration management procedures and structures to be used.
Maintenance plan	Predicts the maintenance requirements of the system, maintenance costs and effort required.
Staff development plan	Describes how the skills and experience of the project team members will be developed.

Project Planning Process

Establish the project constraints

Make initial assessments of the project parameters

Define project milestones and deliverables

while project has not been completed or cancelled loop

 Draw up project schedule

 Initiate activities according to schedule Wait (for a while)

 Review project progress

 Revise estimates of project parameters Update the project
 schedule

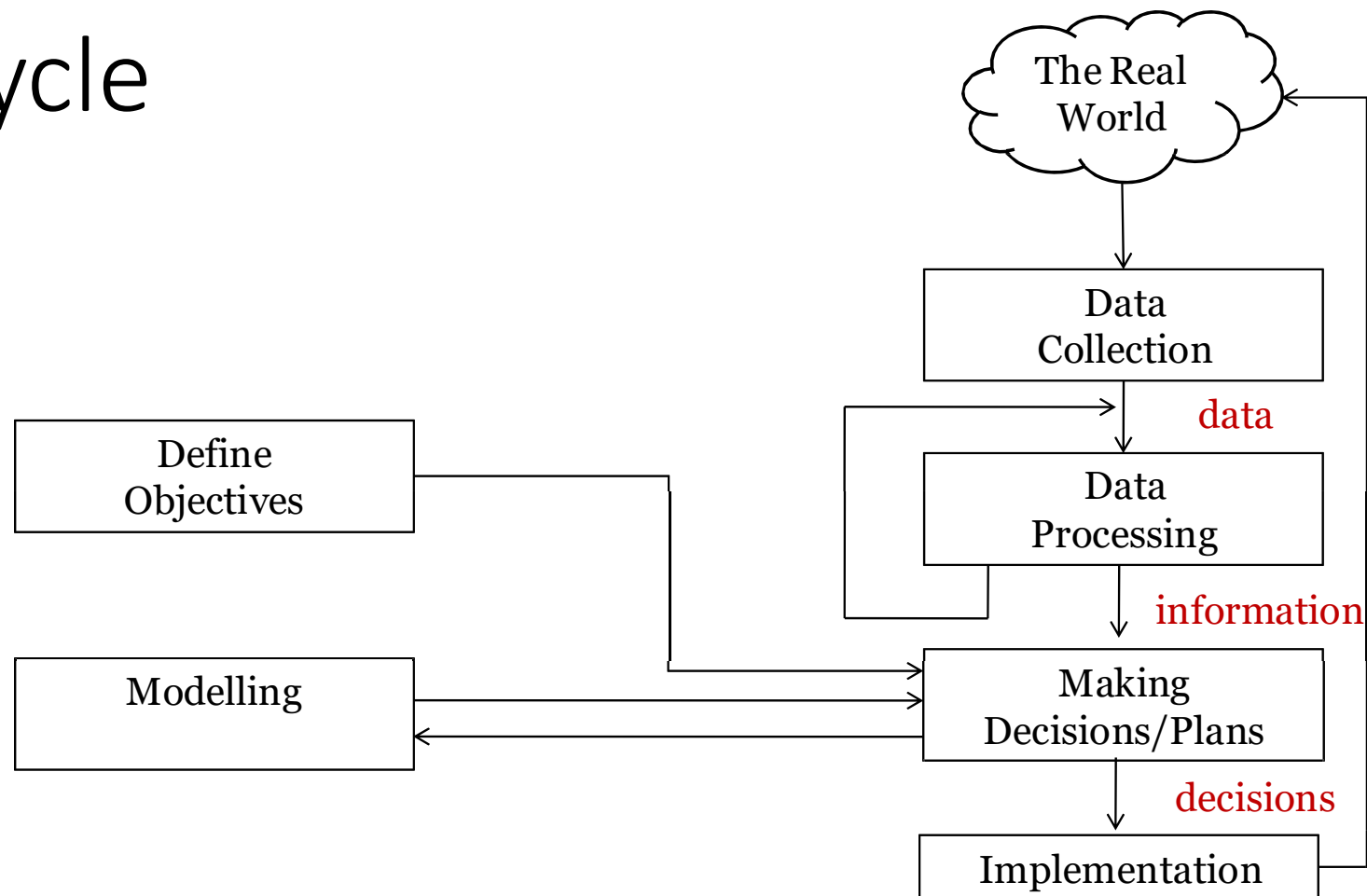
 Re-negotiate project constraints and deliverables if (problems
 arise) then

 Initiate technical review and possible revision

 end if

end loop

The Project Control Cycle



Definition of Software Project Management

Software Project Management is the process of planning, executing, and monitoring software projects to achieve their goals effectively and efficiently.

Importance of managing software projects effectively

Software projects are critical in today's technology-driven world, and successful management is crucial for meeting objectives.

Characteristics of a Good Project Manager:

- A good project manager is a key factor in the success of any project, including software projects. Here are some characteristics that define a good project manager:
1. **Leadership Skills:** A good project manager is a strong leader who can guide the project team, make critical decisions, and motivate team members to achieve their best.
 2. **Effective Communication:** Clear and concise communication is essential in project management. A skilled project manager can convey complex technical information to both technical and non-technical team members and stakeholders.
 3. **Problem-Solving Ability:** Projects often encounter unexpected issues and challenges. A good project manager is adept at identifying problems, analyzing them, and finding effective solutions.
 4. **Adaptability:** The software industry is dynamic, and requirements can change rapidly. A successful project manager is adaptable and can manage evolving project scopes and priorities.
 5. **Technical Proficiency:** While not always required, a solid understanding of software development processes can greatly aid a project manager in making informed decisions and understanding the team's work.
 6. **Organizational Skills:** Project managers must keep track of various tasks, timelines, and resources. Being organized and able to manage these efficiently is crucial.
 7. **Risk Management:** Identifying and managing risks is a vital part of project management. A good project manager assesses potential risks and develops strategies to mitigate them.
 8. **Team Management:** The ability to lead and support a project team is vital. A good project manager fosters teamwork, resolves conflicts, and helps team members perform at their best.
 9. **Time Management:** Projects have deadlines and schedules. A skilled project manager knows how to manage time effectively to ensure the project stays on track.
 10. **Stakeholder Management:** A successful project manager understands the importance of managing and meeting the expectations of stakeholders, such as clients, sponsors, and end-users.

Successful Software Project Manager:

- A successful software project manager is one who consistently achieves project goals and delivers high-quality software. Here are some qualities and abilities that contribute to their success:
1. **Effective Risk Management:** Successful project managers proactively identify and manage risks to prevent issues from derailing the project.
 2. **Strong Stakeholder Engagement:** They actively involve and engage stakeholders to ensure their needs and expectations are met throughout the project.
 3. **Motivational Leadership:** Successful project managers inspire and motivate their team members, creating a positive and productive working environment.
 4. **Technical Proficiency:** A deep understanding of software development processes and technology trends allows them to make informed decisions and guide the team effectively.
 5. **Adaptability and Agile Thinking:** They are flexible and can adapt to changing project requirements and circumstances, often leveraging Agile methodologies to their advantage.
 6. **Effective Problem-Solving:** They are skilled at identifying problems and finding creative solutions to overcome challenges.
 7. **Clear Communication:** Successful project managers excel in communication, ensuring that everyone on the team understands their roles and responsibilities.
 8. **Quality Focus:** They prioritize delivering high-quality software by implementing thorough testing and quality assurance processes.
 9. **Scheduling and Time Management:** They create realistic project schedules and efficiently manage time and resources.
 10. **Continuous Improvement:** Successful project managers are committed to ongoing improvement, learning from each project and applying lessons to future endeavors.

Overview of Software Project Planning:

- Software project planning is the initial phase of a software project where the project's objectives, scope, and requirements are defined, and a plan is developed to guide the project to completion. Here's an overview of software project planning:
- 1. Defining Project Scope:** The first step is to determine the project's boundaries, objectives, and what is considered "in scope" and "out of scope."
 - 2. Setting Clear Objectives:** Clear and measurable project objectives are established. These objectives serve as a roadmap for the project.
 - 3. Resource Allocation:** Identify and allocate the necessary resources, including human resources, equipment, and tools, to carry out the project.
 - 4. Creating Project Schedules:** Develop a detailed project schedule that outlines the sequence of tasks, milestones, and deadlines.
 - 5. Risk Assessment:** Identify potential risks that may impact the project and develop strategies to mitigate these risks.
 - 6. Budgeting:** Determine the project's budget and allocate funds for various project activities.
 - 7. Quality Assurance and Testing Planning:** Plan for quality control and testing processes to ensure the software meets specified quality standards.
 - 8. Team Roles and Responsibilities:** Define the roles and responsibilities of each team member, clarifying who does what throughout the project.
 - 9. Communication Plan:** Establish a communication plan to ensure all team members and stakeholders are informed about project progress and changes.
 - 10. Documentation:** Ensure that project documentation, including requirements, design, and testing plans, is created and maintained throughout the project.
- Software project planning is a crucial phase that sets the foundation for successful project execution. A well-planned project is more likely to stay on schedule, within budget, and meet or exceed its objectives.